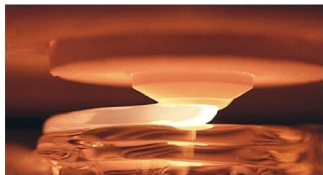


## MIT unveils 3D printing of glass objects

The field of 3D printing is still largely the domain of major companies in need of easier prototyping methods and hobbyists. But a new breakthrough could vastly expand the technology's applications, that is, 3D printed glass.



3D printing of glass

MIT has unveiled a method it calls G3DP that allows the creation of complex 3D glass structures to be printed in a similar fashion to plastic constructs.

MIT's process accomplishes this by using two chambers, one that acts as a kiln cartridge and another that works to melt the structures together. The molten glass is distributed through an alumina-zircon-silica nozzle that pours the material out like soft-serve ice cream.

The G3DP process was developed by a team including MIT Media Lab's Mediated Matter group, MIT Glass Lab, Wyss Institute and MIT's Mechanical Engineering Department.

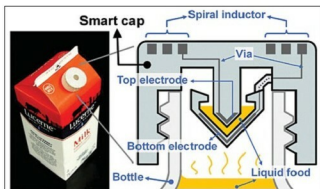
One of the project's researchers, Neri Oxman, says that this could also lead to advances in creating fibre-optic cables that transmit data more efficiently.

## 3D-printed smartcap to sense spoiled food

UC Berkeley (USA) engineers, in collaboration with colleagues at Taiwan's National Chiao Tung University, are expanding the already impressive portfolio of 3D printing technology to include electrical components. They have put the new technology to test by printing a wireless smartcap for a milk carton that can detect signs of spoilage using embedded sensors.

Polymers and other such materials are poor conductors of electricity, and thus bad candidates for electronic devices. To get around this, the researchers started off by building a system using polymers and wax. They then removed the wax, leaving hollow tubes into which liquid metal—in their experiments they used silver—was injected and then cured.

Researchers integrated the electronic components into a plastic milk carton cap to monitor signs of spoilage. The smartcap was fitted with a capacitor and an inductor to form a resonant circuit. A quick flip of the carton



UC Berkeley engineers created a smartcap using 3D-printed plastic with embedded electronics to wirelessly monitor the freshness of milk (Photo and schematic by Sung-Yueh Wu)

allowed a bit of milk to get trapped in the cap's capacitor gap, and the entire carton was then left unopened at room temperature for 36 hours.

The circuit could detect the changes in electrical signals that accompany increased levels of bacteria. The researchers periodically monitored the changes with a wireless radio-frequency probe at the start of the experiment and every 12 hours thereafter, up to 36 hours.

## Facebook's solar-powered drone to beam Internet from the sky

Facebook has completed the production of its first full-scale solar-powered Internet drone that will deliver wireless Internet with lasers from the sky in parts of the developing world where there is a lack of connectivity.



Facebook's solar-powered drone to beam Internet from the sky

Mark Zuckerberg, co-founder, Facebook, has announced the completion of Aquila, Facebook's first unmanned plane that beams down Internet connectivity from the sky, as part of his project with *Internet.org*.

Aquila's wings are made of carbon fibre. It has the wingspan of a Boeing 737, but weighs less than a car and can stay in the air for months at a time.

Facebook will have lasers on the ground that can locate the dome-shaped optical head, located on the bottom of the plane, in the air, basically shooting a laser at a dime-sized target that is more than 16km (10-miles) away. The plane will first hone in on the general location of the laser on the ground, proceeding to target it further and lock onto the location so that it can start beaming down the Internet.